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Bay-capping reactions: Kinetics and influence on graphene-edge growth

Xiaoqing You [a,*], Russell Whitesides [b], Dmitry Zubarev [c], William A. Lester Jr. [c,d], Michael Frenklach [a,e]

[a] Department of Mechanical Engineering, University of California, Berkeley, CA 94720-1740, USA
[b] Lawrence Livermore National Laboratory, Livermore, CA 94550, USA
[c] Kenneth S. Pitzer Center for Theoretical Chemistry, Department of Chemistry, University of California, Berkeley, CA 94720-1460, USA
[d] Chemical Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA
[e] Environmental Energy Technologies Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA

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Abstract

A capping reaction was found in a recent study to cause curvature in graphene edges and affect their growth rates. This reaction is initiated by acetylene addition to an armchair-like graphene radical site containing an embedded five-member ring. In this work, a combination of quantum-chemical calculations and reaction rate analysis was employed to explore the elementary steps of bay-capping reactions and to derive their rate coefficients. To evaluate the influence of the embedded five-member ring on six-member ring cyclization, the energetics and reaction rates of capping an embedded five-member ring were compared with those of capping a graphene armchair site. Detailed kinetic Monte Carlo simulations were performed to model evolution of graphene edges at conditions typical of soot growth and to examine the influence of the newly-derived rate coefficients.
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1. Introduction

Controlling formation of soot in modern combustion systems requires a good mechanistic understanding of soot particle growth. One of the biggest unknowns, particle inception, received substantial attention in recent years [1–13] and its mechanistic picture steadily advances toward a unified theory [1,13]. Another part of the mechanism, surface growth, has transformed from an empirical rate law [14] to a simple elementary-reaction model [15], to a complex, multi-step process [9,10,16–19]. The complexity of the surface growth mechanism stems from numerous possibilities of chemical transformations taking place at graphene edges, presumed to be the building units of soot particles. Among them are game-changing processes: migrations of lone [17] and embedded [18,20] five-member rings, which give rise to rapid lateral translation of five-member rings along zigzag edges.

Most recently, Whitesides and Frenklach [18] presented a new detailed kinetic Monte-Carlo (KMC) model of graphene-edge growth with a total of 42 surface transformations. Evolving edge

* Corresponding author.
E-mail address: xiaoqing.you@berkeley.edu (X. You).

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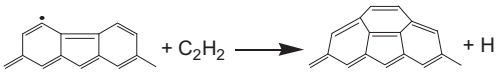
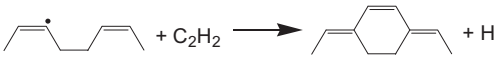
morphology and growth rate were found to be affected by the rates of key reactions. One of the most interesting among them is capping, addition of acetylene to an embedded five-member ring,



(R1)

The significance of this reaction step is several-fold [18]. First, it pins down the five-member ring and prevents it from migrating. As ring migration generally leads to smoother surfaces, this aspect of capping works against it. Second, capping of five-member rings forms armchair-like sites that initiate zipper-type growth of new layers on graphene edges. Finally, capping permanently embeds five-member rings into the growing graphene structure and hence causes it to curve.

In the Whitesides and Frenklach study [18], reaction R1 was assumed to have the rate coefficient of its zigzag-edge analog because no rate data were available. The importance of reaction R1 calls for a thorough examination of its kinetics, the subject that motivated the present study. Capping an embedded five-member ring looks very similar to the acetylene-addition step of the primary HACA sequence [1], the one that takes place on an armchair bay,



(R2)

We hence set out to compare the rate coefficients of the two reactions and their influence on graphene-edge growth. A graphene sheet becomes curved during ring-closure in capping an embedded five-member ring on a zigzag edge, while it remains planar when capping a boat site on an armchair edge.

Re-evaluation of reaction R2 turned out to be important in its own right. In the KMC simulations of Ref. [18], the rate coefficient for the reaction R2 (reaction 27 in Ref. [18]) was taken from a prior KMC study [17]. The latter originated from Appel et al. [21] who assumed, following the discovery of a faster ring-closure pathway initiated by H migration [22], that the rate of R2 to be twice the high-pressure limit of acetylene addition to phenyl computed by Wang and Frenklach [23]. The two prior theoretical studies of this reaction [22,23] were performed at semi-empirical levels of quantum theory, PM3 and AM1, respectively. In the present study, we reexamine the kinetics of the complete system of reaction R2, with both the initial [23] and the new [22] pathways considered simultaneously. The calculations were performed at a higher level of theory, density functional theory (DFT), the same employed in recent studies of similar-size systems [9,19,20,24–26]. To examine the effect of substrate size, we performed the calculations for reaction R2 using two substrates, 3- and 6-ring molecular structures. The three reactions examined are shown in Fig. 1.

2. Computational methodology

2.1. Energetics

In DFT calculations the B3LYP hybrid functional [27] and the 6-311G(d, p) basis set were employed to optimize geometries and calculate vibrational frequencies for all the stationary points on the reaction pathways. Transition states were confirmed to connect the reactant and product stable species by visual inspection of the normal modes of the corresponding imaginary frequencies. The potential energies computed at the B3LYP/6-311G(d, p) level for stable species of the size studied here were reported to be in good agreement with experimental and high-level *ab initio* results [28,29]. Underprediction of transition state energies by this method of about 5 kcal mol^{-1} has been reported [30,31], which results in rate coefficients with order-of-magnitude accuracy. Zero-point energies (ZPE) and vibrational frequencies were scaled by a factor of 0.967 [32]. All the *ab initio* calculations were carried out using the Gaussian 03 program [33].

It is pertinent to mention that the initial assignment of the rate coefficient of the capping reaction, R1, was done (in Ref [18]) entirely "by analogy", as no data could be found. To our knowledge, the present work is the first one to undertake a detailed theoretical study for this reaction. During the course of the study, we took a look at a related capping reaction, R2. The rate coefficient for the latter, which has propagated through the literature, originated in our laboratory some time ago. Re-evaluation at a higher level of theory possible today produced an order-of-magnitude lower value, which is significant even in light of the 5 kcal mol^{-1} uncertainty of the method.

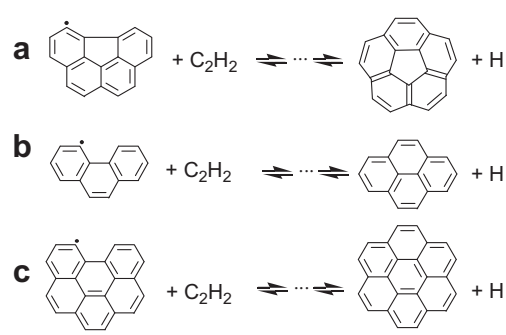


Fig. 1. Reaction systems analyzed.

2.2. Kinetics

The kinetics of the three chemically activated reaction systems were examined using the MultiWell suite of codes [34,35]. The key inputs—reaction barriers, frequencies, and moments of inertia – were assigned from DFT calculations at the B3LYP/6-311G(d, p) level of this study. MultiWell simulations were performed for temperatures ranging from 1500 to 2500 K and pressures ranging from 0.1 to 10 atm, spanning conditions from typical laboratory studies to practical combustion devices. Lennard–Jones parameters for the intermediates were estimated from an empirical correlation [36]. Argon was chosen as the bath gas collider. The collisional energy transfer was treated using the exponential-down model with $\langle \Delta E_{down} \rangle = 260$ cm^{-1} [37]. The density and sum of states were determined by exact count with an energy grain size of 10 cm^{-1}, and a maximum energy of 500,000 cm^{-1}, so that the high-pressure limit rate coefficients calculated from the density and sum of states were almost equal to those calculated from canonical transition-state theory. The numerical runs were carried out for reaction times ranging from 10^{-7} to 10^{-5} s, long enough for the system to reach equilibrium. For each set of conditions, 10^6–10^7 stochastic trials were performed to maintain statistical error within 1–4% for species mole fractions.

2.3. Kinetic Monte Carlo simulation

The effects of the newly-derived rate coefficients were tested in detailed KMC simulations performed using the model and code of Whitesides and Frenklach [18]. The basic idea of the simulations is to track the evolution of a graphene structure in gaseous environments typical of combustion, yet with fixed properties, e.g., temperature and species compositions, to be able to discern the influence of individual factors. At each time step, a reaction event was selected stochastically from a total of 42 surface transformations and applied following the Gillespie algorithm [38,39]. To properly account for the evolving curvature, after application of the KMC step, the molecular structure was optimized using the MM3 potential [40] with the TINKER molecular mechanics package [41]. A check was made after each geometry optimization to ensure that the geometry produced by the molecular mechanics code was consistent with the bonding implied by the KMC model. Further computational details can be found in Refs. [18,42].

3. Results and discussion

3.1. Reaction energetics

We explored the potential energy surfaces at the B3LYP/6-311G(d, p) level for the three reaction systems $C_{18}H_9 + C_2H_2$, $C_{14}H_9 + C_2H_2$, and $C_{22}H_{11} + C_2H_2$, denoted hereafter as systems **a**, **b**, and **c**, respectively (see Fig. 1). The computed relative energies of the reaction species are summarized in Table 1; a detailed report of the computed molecular properties for the species involved – electronic energies, geometries, and vibrational frequencies – is given in the Supplemental material (Tables S1 and S2).

Comparison of the computed potential energy surfaces indicates that the three systems have mechanistically similar reaction pathways; that of system **a** is displayed in Fig. 2. Taking the reaction pathway of system **a** as an example, we note the following mechanistic features. The minimum energy path starts by C_2H_2 addition to the reactant radical, **1**, forming a chemically activated radical, **2**. This chemically activated radical can decompose back to the reactants, be stabilized via collisions with the third body, lose a hydrogen atom to generate stable species **6**, isomerize to form species **3**, or undergo ring-closure to generate species **4**. Species **5** is generated from species **3** via ring-closure or from **4** by a hydrogen-shift reaction, and finally the product, **7**, is produced by losing a hydrogen atom from species **4** and **5**. The reactions steps of systems **b** and **c** are similar to **a**, except in **b** and **c** no transition state was found for the isomerization between **4** and **5**. This may be due to the planar structure of **4** and **5** in **b** and **c** with shorter bond length of the two carbon atoms involved, which makes dissociation of H atom more likely than a shift to the neighboring C-atom. In contrast, the curvature induced by the embedded five-member ring in system **a** makes the isomerization feasible.

Inspection of the results in Table 1 and Fig. 2 shows that systems **b** and **c** are similar when comparing overall reaction enthalpy, barrier heights,

Table 1
Energies of species in reaction systems (kcal mol^{-1}) from B3LYP/6-311G(d, p) including scaled ZPE.

Species	Reaction system		
	a	**b**	**c**
1 + C_2H_2	0.0	0.0	0.0
2	−38.5	−26.1	−26.5
3	−37.2	−33.4	−34
4	−45.3	−61.9	−68.4
5	−60.5	−82.2	−85.6
6 + H	−4.6	2.9	2.4
7 + H	−26.4	−51.6	−57.6
TS 1–2	3.7	6.8	6.6
TS 2–3	−23	−23.6	−24.1
TS 2–4	−16.5	−23.5	−24.6
TS 2–6	0.2	7.9	7.3
TS 3–5	−22.4	−29.1	−29.6
TS 4–5	−18.1		
TS 4–7	−21.8	−43.6	−49.7
TS 5–7	−24.8	−49.2	−54.9

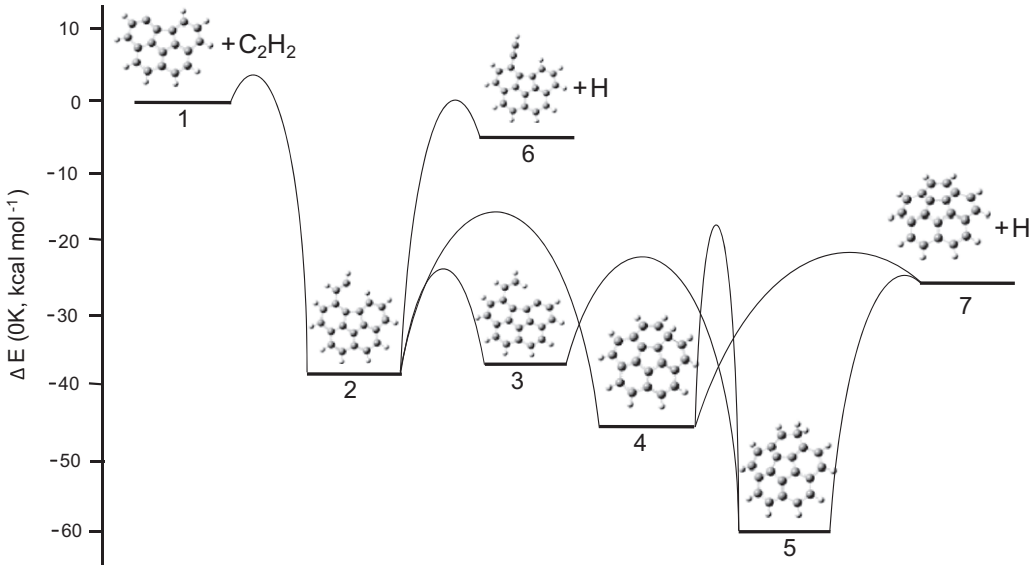


Fig. 2. Minimum energy path for capping an embedded five-member ring (system **a**).

and the relative energy of individual species. However, system **a** is quite different from **b** and **c** in that respect. For instance, the reaction $\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$ of system **a** is less exothermic by 25–31 kcal/mol than systems **b** and **c**, and the relative energies of the curved intermediates **4** and **5** of **a** are 17–25 kcal/mol higher than those of **b** and **c**. As expected [43], the five-member ring embedded in the structure not only causes curvature, but also leads to a higher potential energy.

3.2. Reaction kinetics

Using the results of the quantum–chemical calculation, we computed rate coefficients at a range of temperatures, 1500–2500 K, and pressures, 0.1–10 atm. No pressure dependence was found for the rate coefficients of all three reaction systems. The dominant products were corannulene, pyrene, and coronene in systems **a**, **b**, and **c**, respectively, as summarized in Fig. 1.

Rate coefficients computed at 1500, 2000, and 2500 K for individual reaction steps are listed in Table 2. As can be seen from these results, at all three temperatures the isomerization reactions of the intermediates are fast and the dissociation rates of the intermediates are too large to allow any stabilization. Comparing systems **b** and **c** indicates that the rate coefficients of all the corresponding reaction steps are within a factor of 3 from each other. Similar to and caused by the differences in the energetics, the rate coefficients of some reaction steps of system **a** are much larger than the corresponding steps of **b** and **c**.

To compare systems **a** and **c** further, the rate coefficients computed for the main reaction channels, $\mathbf{1} + C_2H_2 \rightarrow \mathbf{6} + H$ and $\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$, are displayed in Fig. 3. This comparison shows that the rate coefficient of the step generating $\mathbf{7} + H$ of **a** is about twice that of **c**. The rate coefficient to produce $\mathbf{6} + H$ has a negative temperature dependence for system **a**, but a large positive temperature dependence for **c**. This difference is largely due to the difference in the barrier height of the transition state, **TS 2–6**, which is much higher for **c** than it is for **a**.

3.3. KMC simulations

Using the rate coefficients computed for systems **a**, **b**, and **c**, we updated three of the 42 surface transformations of the Whiteside and Frenklach model [18], reactions 27, 35, and 36, as shown in Table 3. Considering that a larger substrate should be a better representation of a soot surface, we chose the rate of reaction channel $\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$ of the larger system, **c**, for KMC reaction 27. KMC reaction steps 35 and 36, were assigned rate coefficients of reaction channels $\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$ and $\mathbf{6} + H \rightarrow \mathbf{7} + H$ of system **a**, respectively. Other reaction channels from the reactants $\mathbf{1} + C_2H_2$ to intermediates of systems **a**, **b**, and **c** were computed to be negligibly slow and hence were not included in the KMC simulations.

The new rate coefficients are compared to the previous ones in Fig. 4. Compared to the old values, the new k_{36} is slightly lower within essentially the entire temperature range of the present study, and the new k_{35} is higher at high temperatures and lower at low temperatures. The largest difference

Table 2
Rate coefficients (units: cm^3, mol, sec) for elementary reactions of systems **a**, **b**, and **c**.

Reaction	**a** Temperature (K)			**b** Temperature (K)			**c** Temperature (K)		
	1500	2000	2500	1500	2000	2500	1500	2000	2500
1 + C_2H_2 → **2**	2.34×10^{12}	6.43×10^{12}	1.34×10^{13}	1.28×10^{11}	4.44×10^{11}	1.06×10^{12}	1.26×10^{11}	4.32×10^{11}	1.03×10^{12}
2 → **1** + C_2H_2	1.86×10^{9}	8.07×10^{10}	7.74×10^{11}	6.27×10^{9}	1.22×10^{11}	7.27×10^{11}	6.05×10^{9}	1.2×10^{11}	7.22×10^{11}
2 → **3**	1.56×10^{10}	6.37×10^{10}	1.5×10^{11}	1.7×10^{12}	2.39×10^{12}	2.98×10^{12}	3.85×10^{12}	5.4×10^{12}	6.7×10^{12}
2 → **4**	3.06×10^{8}	1.97×10^{9}	6.03×10^{9}	4.28×10^{11}	5.45×10^{11}	6.32×10^{11}	1.09×10^{12}	1.3×10^{12}	1.45×10^{12}
2 → **6** + H	4.8×10^{8}	1.69×10^{10}	1.45×10^{11}	1.54×10^{9}	3.62×10^{10}	2.45×10^{11}	1.41×10^{9}	3.26×10^{10}	2.18×10^{11}
3 → **5**	3.76×10^{9}	1.32×10^{10}	2.81×10^{10}	5.15×10^{11}	7.54×10^{11}	9.49×10^{11}	6.11×10^{11}	9.06×10^{11}	1.15×10^{12}
4 → **5**	1.98×10^{9}	2.32×10^{10}	1.02×10^{11}						
4 → **7** + H	1.71×10^{10}	1.58×10^{11}	6.07×10^{11}	4.89×10^{10}	2.85×10^{11}	8.34×10^{11}	4.88×10^{10}	2.96×10^{11}	8.83×10^{11}
5 → **7** + H	7.77×10^{8}	2.05×10^{10}	1.47×10^{11}	8.72×10^{8}	1.8×10^{10}	1.12×10^{11}	1.96×10^{9}	3.32×10^{10}	1.84×10^{11}

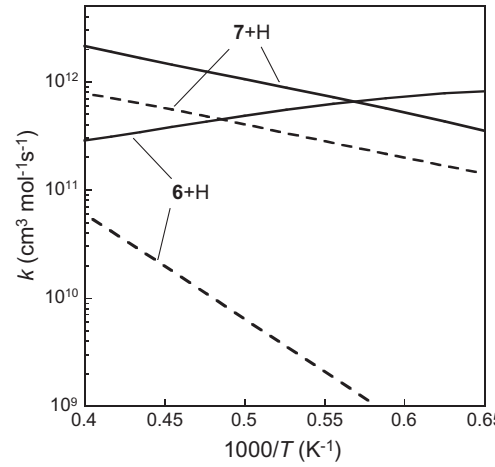


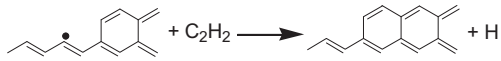
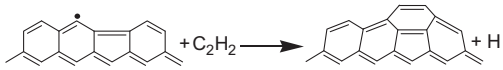
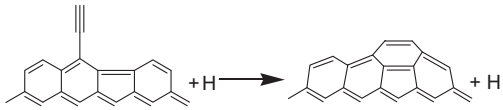
Fig. 3. Rate coefficients of reactions **1** + C_2H_2 → **6** + H and **1** + C_2H_2 → **7** + H for system **a** (solid lines) and system **c** (dashed lines).

is found for the rate coefficient of KMC reaction 27: the new k_{27} is about one order of magnitude lower than the previous one. The latter result turns out to have the largest influence on the graphene-edge growth as is discussed next.

To assess the influence of the changes in the rate coefficients, we performed a series of KMC simulations, replicating the conditions of the previous study [18]. The simulations were run on five different substrates: pyrene, coronene, pentacene, decacene, and icosacene, at three constant temperatures 1500, 2000, 2500 K, and a constant gas-phase composition with mole fractions of C_2H_2, H_2, and H set to 0.1, 0.1, and 0.01, respectively. The numerical results of these simulations were analyzed using the following measures [18]: *added C-atoms*, the number of carbon atoms added to the evolving molecular structure; *growth rate*, the slope of added C-atoms at a given time; f_{R5}, the fraction of five-member rings among all carbon rings of the evolving molecular structure; *number of edge bifurcations*, a permanent discontinuity in the developing graphene edge, defined as six or more solid carbon atoms in succession of which at least one is part of a five-member ring; and f_{AC}, the fraction of surface sites that can accommodate cyclization by addition of C_2H_2. In light of the stochastic nature of the simulations, the above measures were averaged over a series of 15 simulations runs.

The replacement of the three reactions rates produced, generally, similar effects on the growth rates for the five substrates studied. Compared with the previous results [18], C-atom addition remained nearly the same at 1500 K, but slowed down at 2000 and 2500 K, and especially at 2000 K. As an example, Fig. 5 displays addition of C atoms in KMC simulations started with

Table 3
Updated rate coefficients of three KMC reaction steps.

No. in Ref. [18]	Surface reaction step	Elementary reaction of the present study	Rate coefficient k ($cm^3\ mol^{-1}\ s^{-1}$)
27		$\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$ (**c**)	$1.9 \times 10^{10} T^{0.712} e^{-4893/T}$
35		$\mathbf{1} + C_2H_2 \rightarrow \mathbf{7} + H$ (**a**)	$1.0 \times 10^{10} T^{0.955} e^{-5326/T}$
36		$\mathbf{6} + H \rightarrow \mathbf{7} + H$ (**a**)	$7.0 \times 10^{10} T^{0.018} e^{6059/T}$

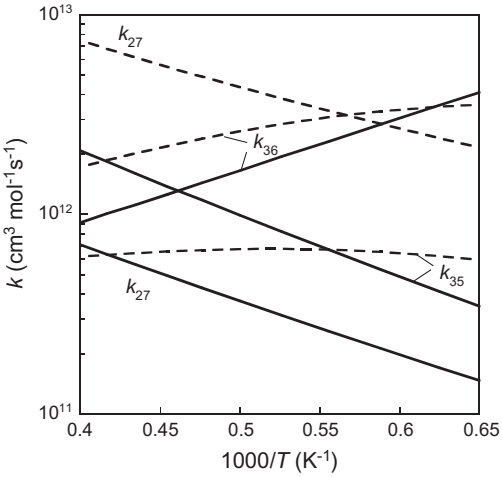


Fig. 4. Rate coefficients of KMC reaction steps: solid lines – this study, dashed lines – Ref. [18].

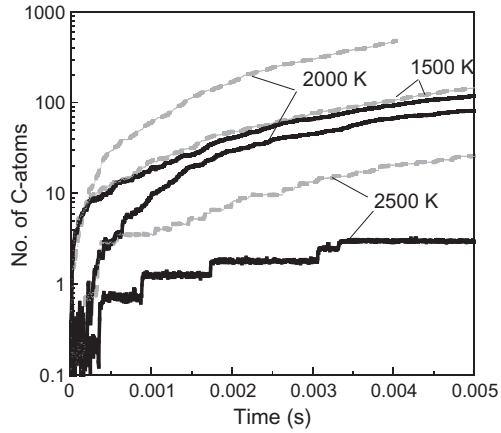


Fig. 5. Addition of C-atoms for substrate coronene: black lines – this study, grey lines – Ref. [18].

coronene, comparing the results obtained before and after changing the rate coefficients of reactions 27, 35, and 36. Before the change, growth is the fastest at 2000 K, and the slowest at 2500 K. After the change, growth at 2500 K remains the slowest, but growth at 2000 K is now slower than that at 1500 K. Performing KMC simulations with changing rates individually revealed that the difference in the growth-rate pattern is mainly due to the change in the rate of KMC reaction 27.

Among other computed measures, f_{R5}, the fraction of five-member rings, increased from 0.24% to 1.01% at 0.004 s for coronene growth at 2000 K because the number of six-member rings was significantly decreased after lowering k_{27}. Other than that, only f_{AC}, the fraction of sites that can accommodate cyclization by addition of C_2H_2, was appreciably affected. As shown in Fig. 6, there is no appreciable change in f_{AC} at 1500 K. However, the values of f_{AC} at 2000 and 2500 K are smaller after changing the rate coefficients of KMC reaction steps 27, 35, and 36. The lower k_{27}, as discussed above, reduces the graphene-edge growth, implying that the surface becomes less reactive as it has a smaller number of cyclization-enabling sites.

The results obtained here on the growth rate, f_{R5}, and f_{AC} emphasize the importance of the key reaction rates. Previous KMC studies [18] showed that the difference in the growth rate and morphology of the evolving graphene edge was due to competition among corner, interior, and embedded-ring nucleation events initiating zipper-type growth. The mechanism of graphene-edge growth changes with temperature. At 1500 K, graphene-edge growth is promoted by the stable five-member rings incorporated in the growing layer that create interior nucleation sites.

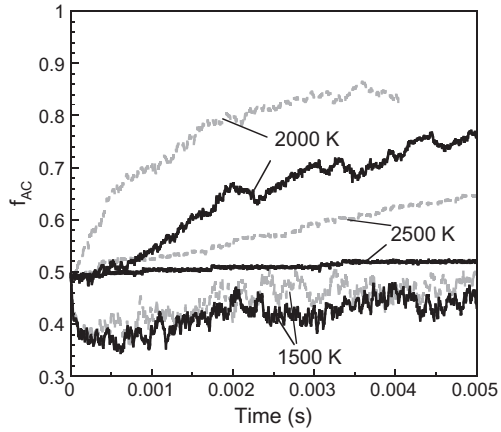


Fig. 6. f_{AC} for substrate coronene: black lines – this study, grey lines – Ref. [18].

Embedded-ring nucleation via capping the embedded five-member ring, KMC steps 35 and 36, is also possible at this temperature. However, the new rates of KMC reactions 35 and 36 do not change significantly; therefore, the growth rate, f_{R5}, and f_{AC} at 1500 K are not appreciably affected by changing the rates. With the increase in temperature, the thermodynamic and kinetic stability of five-member rings decreases. The increasingly frequent desorption of five-member rings leaves the structure composed of predominantly six-member rings, which promotes the relative importance of boat-site cyclization, KMC reaction 27. Decreasing its rate tenfold has the most dramatic effect at 2000 K to reduce the growth rate and the value of f_{AC} but to increase the value of f_{R5}. At 2500 K, growth becomes much less frequent and five-member rings are rarely incorporated into the structure. The reaction mechanism at this temperature is dominated by corner nucleation via formation of six-member rings; and hence lowering the rate of KMC reaction 27, which propagates edge growth, causes a smaller reduction in the overall growth rate.

Representative snapshots of KMC simulations are displayed in Fig. S1 in Supplemental data. The molecular structure formed at 1500 K is significantly curved because of the high fraction of five-member rings in the structure. As temperature rises, the embedded five-member rings are more likely to escape from the surface; therefore, the resulting structure is less curved at 2000 K, and stays planar at 2500 K without much growth. It is noteworthy that while the change in the rate coefficients of the three reactions lowered the overall growth rates, it did not affect the phenomenon of curving obtained for graphene-edge evolution in the previous study [18].

Experimental evidence can be found for the presence of graphene sheet curvature (e.g., [44,45]). However, our attempts to explore possible correlations between the present numerical results and available experimental observations produced no sufficient evidence for making comparison in this case aside from general speculations.

4. Summary

A recent study [18] called for a thorough examination of bay-capping reactions because of their importance to the morphology and rate of graphene-edge growth. We employed a combination of quantum-chemical calculations and reaction rate analysis to explore the elementary steps of these types of reactions and to derive their rate coefficients. To evaluate the influence of the embedded five-member ring on six-member ring cyclization, we compared the energetics and reaction rates of capping an embedded five-member ring with those of capping a "regular" armchair graphene-edge site.

The rate coefficient calculated here for capping the *five*-member-ring bay is of the same order of magnitude as its prior estimate, yet with a substantially different activation energy. The calculated rate coefficient for capping the *six*-member-ring bay is one order magnitude lower than that used in prior KMC studies and is close to the older evaluation [23].

To examine the influence of the new kinetic data, we performed detailed kinetic Monte Carlo simulations changing the rate coefficients of the bay-capping reactions of the initial model. With the new rate coefficients, the overall growth rates of graphene edge were lower. The most dramatic effect was obtained at 2000 K. At the same time, the change in the rate coefficients did not affect the phenomenon of curving obtained for graphene-edge evolution in the previous study [18].

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.110.

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